

Early Disease Detection using Android Application based on Nail's Image

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Abstract—The advancement in medical science has played a major role in disease diagnosis. Due to this advancement, diagnosis of diseases is made simple, and many diseases can be diagnosed referring to the characteristic changes in human eyes, tongue, skin, etc. Similarly, the human nail is also one of the significant disease detectors for diseases such as kidney disease, heart disease, anemia, etc., even in ancient times and also stated in Ayurveda. In this paper, an android application is developed for predicting the disease using nail images. A model is prepared for dataset collection. Two technologies are used during experimentation image detection and prediction using state of the art end-to-end tensor flow platform. To have a more reliable and efficient system in prediction algorithms like SVM, KNN, Random Forest, and CNN tested on the prepared dataset. The comparative study shows that CNN is best suited for the application with a testing accuracy of 75 percent and training accuracy of 98 percent.

Keywords— *Nail image, Android application, TensorFlow, Flutter, Object detection.*

I. INTRODUCTION

The human body comprises many body parts, tissues, organs, and systems such as the digestive system, nervous system, etc. The human body contains 78 body parts, namely the brain, heart, liver, lungs, eyes, kidneys, mouth, nose, etc. Some of these body parts also show changes when humans have some disease. The human nail is divided into parts, namely matrix (root), lunula (half- moon), nail plate, cuticle, and the free edge of the nail, etc. [1]–[3]. Mostly the changes occur in the nail plate and hence acts as the primary part for disease diagnosis. There are many unique features of the nail that can be used to make the diagnosis, the selection of features is up to the requirement. The morphological changes in nail unit i.e., abnormalities in the shape of nail such as clubbing, pincer nail, parrot beak nail give a direct indication of disease, example clubbing may indicate a person being HIV positive [4]. Based on the same feature (shape of the nail) the deformation of the nail can be detected using the strain sensors. Prediction of the further deformation or any abnormalities will be done using the synergic working of algorithms along with dataset available to it and sensors, a wearable band is also developed [5]. The skin deformation and posture is also determined using the photo-reflective sensors [6]. A lot of work is also done using inherent properties of the nail such as determining the velocity of the flow of blood through the nail fold. The blood flow speed is an important

index that reflects the state of microcirculation and makes significant marks in diagnosis [7]. A system to analyze nails for disease prediction is needed because the human eye has limitations for recognizing these changes. There are systems available that can provide the prediction of disease and can do diagnosis. There are many rural remote areas where the hospital is very less or not that advanced. This proposed system may be very helpful as it works on the android platform and nowadays every person has at least a basic smartphone. As our system requires only an image of a nail so it can be operated by laymen also. The motive behind this research is to make such an application that can diagnose disease in humans at an early stage so that the cure for the same is also easy and affordable. In our system, we use nail image as a primary body part for disease diagnosis as nails are the farthest from the heart hence receive the oxygen at last [8]. So, if a human is suffering from any major disease such as skin disease, lung disease, etc. then there is a significant change in shape, color [9] texture is shown by the nail. By analyzing such changes in nails user is ready to make predictions of diseases. In our proposed system we developed an application for predicting the disease by using the nail image which will be available on the android platform. In this application, we mainly use two technologies named object detection and TensorFlow. The image detection is used to have faithful input from the user (i.e. nail image only). Then we used the Tflite model of TensorFlow for the prediction part. We have used CNN algorithms for our machine learning part to make predictions. This tflite model and image detection model were running at the backend of our application and for the user, simple UI (User Interface) is designed so that any layman can also be benefited from this application. The structure of the paper is such that, in section II the proposed methodology is explained followed by experiments and results, in section III and conclusion in section IV.

II. METHODOLOGY

A. System Description

The workflow is divided into two parts: Frontend and Backend. The frontend part includes work related to the GUI (Graphical User Interface) implementation of the application. The user interface is designed simple. Once we start the application, the user has to upload the image to the system on the homepage. This can be done by uploading the image from

the gallery or by taking the picture via camera as shown in Fig. 1. Once it is uploaded, the further process is done by the backend part of the application. After uploading an image, the image detection comes into play it checks the input image is the only nail, if not then the process ends at that moment and the user has to initiate the process again to have a prediction, if uploaded image is correct then the user will get the prediction accordingly with the confidence interval.

During the backend process, Disease prediction is done using the CNN model. The resulting prediction is then displayed on the screen with its probability. In the backend part, the input image first undergoes the nail detection process, if the nail is not detected, then the process ends over there and the user has to initiate it again. If detected then proceeds further to the disease detection algorithm. The backend algorithm (CNN) analyzes the nail image and gives the result accordingly. Then the output is displayed on the screen which also provides an option to either test the input again by providing another image or to exit.

A brief overview of the working of the model is shown in Fig. 1 which represents the application's user interface, which is discussed in detail in the UI implementation section. Fig. 2 represents the backend working i.e., functional part of the application, to build this part of the model we worked on dataset preparation, training and testing of data, experimental setup building which is explained in detail in the below sections.

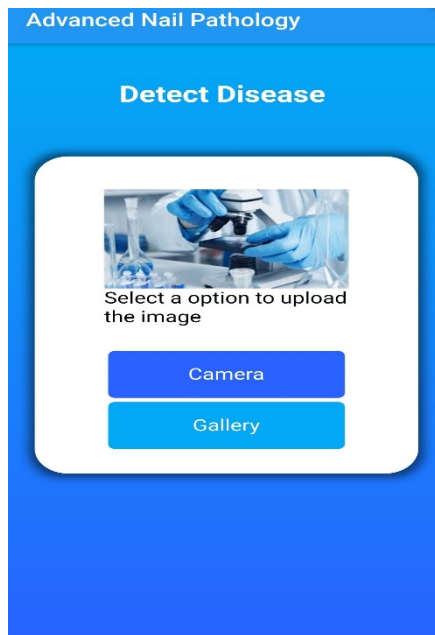


Fig. 1. Frontend View of the Application

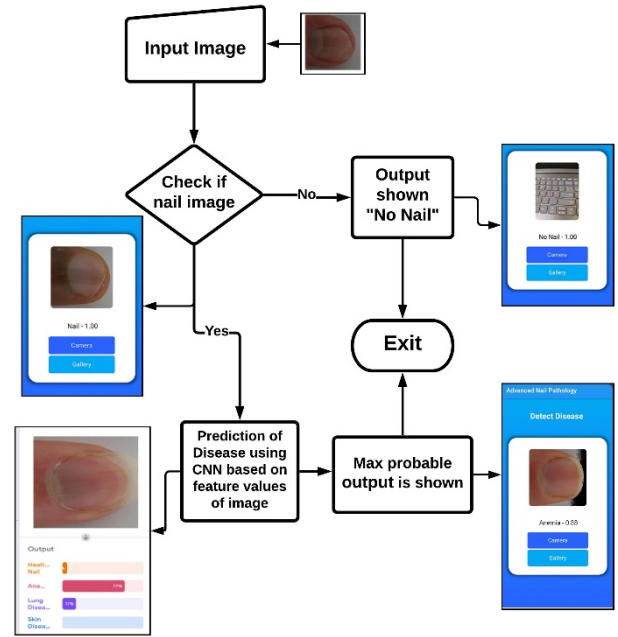


Fig. 2. Backend Working of Model

B. Setup for Dataset Collection

As discussed above we are using red, green, blue and mean pixel values as features for image classification but since all these features are environment-sensitive (light and background) therefore one basic experimental setup is created which provides fixed illumination and background. The setup is shown in Fig. 3 which has an LED array as a light source placed at one extreme of the box and all other surfaces covered with a black sheet to keep the pixel value of the background near to zero, another extreme of the box is used to place the finger beneath the opening provided for placing the camera.

Camera setting used for experimental setup: ISO=200, S=1/90s, WB=6400K, MF= 0.44, EV= -1.17. Users can adjust as per the white balance and Fps of the respective device.

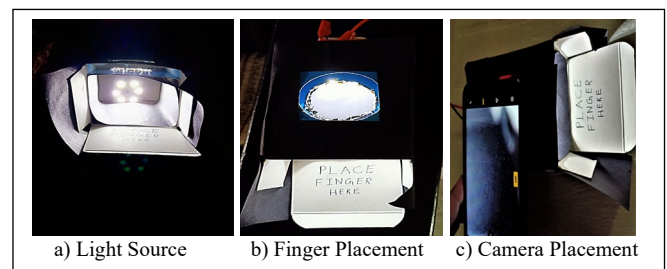


Fig. 3. Image Acquisition Setup

C. Dataset Preparation

Data plays a crucial role in any kind of process such as prediction, analysis, decision making, etc. While predicting, the volume and precision of the dataset should be suitable for the model otherwise the accuracy may go down. The training dataset is made by collecting random nail images which have some disease. Then the image goes through image processing, the image is transformed into its HSV (hue saturation value) equivalent image to stabilize the contrast of the image. Which is further processed by Otsu thresholding, to get clear differences between the region of interest i.e., nail platelet and

surrounding tissues [10] [11]. This process is further processed by extracting the features of the image that is obtained by undergoing the procedure of color Masking and eliminating background by converting it to black pixel to prevent it from participating in feature extraction and giving rise to the misleading prediction by the designed model. The Features extracted were red, green, blue pixel values and the Mean Pixel Value of an image. Example in anemia the symptoms are nail color changes to pale whitish similarly for lungs the nails become yellowish and for skin the disease it is pink or reddish. Based on these features we have trained the SVM model to test other random nail images and classify them according to the class of diseases and prepare a batch of the dataset for implementing CNN for application development. Images used here are from the Naylizer dataset (GitHub), some are taken from the camera (without setup) and others are taken from the setup made (own dataset). So in such a way, we made our heterogeneous dataset to train the CNN model in tf lite as shown in the TABLE I. Images with sr. no 1 to 4 are from Naylizer dataset and sr. no 5 to 6 are from our own dataset, these are some of the images from the whole dataset shown for better understanding of dataset and feature extraction process.

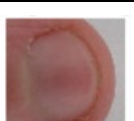
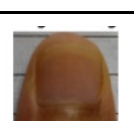
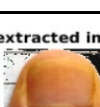
III. EXPERIMENT DETAILS AND RESULTS

A. UI Implementation

The system focuses on providing a better experience to users therefore it is essential to have a user- friendly interface making it more interactive. Therefore, for UI development and backend process integration, a unified environment was used which provides integrated IDE and the fastest tools for developing Android Applications. It is also imperative for the system to be used on every possible platform so that it will help the users to get diagnosed at the early stages of the disease.

So, the system is developed using Flutter SDK (software development kit) which uses Dart programming language for UI development and was implemented with the help of android studio. Now for taking input images two options are available: camera and gallery as shown in Fig. 4a, the user is required to input an image at their convenience. Before disease prediction, the system needs to check if the input image consists of a human nail or not. So, the system needed another ML model which must detect nail images; if not the system should raise output with "no nail" as shown in Fig. 4b and Fig. 4c.

TABLE I. EXTRACTED FEATURES AND RESPECTIVE DISEASES

Sr. no	Input Image	Processed Image			Image Features				Disease
					Red Pixel	Green Pixel	Blue Pixel	Mean Pixel	
1					174.000	32.000	24.000	127.513	Anemia
2					223.000	117.000	69.000	128.898	Healthy nail
3					235.000	130.000	93.000	130.475	Skin Diseases
4					227.000	138.000	117.00	131.086	Lung Disease
5					28.000	28.000	28.000	129.309	Melanonychia
6					166.00	166.00	158.00	128.354	Oxygen deprivation

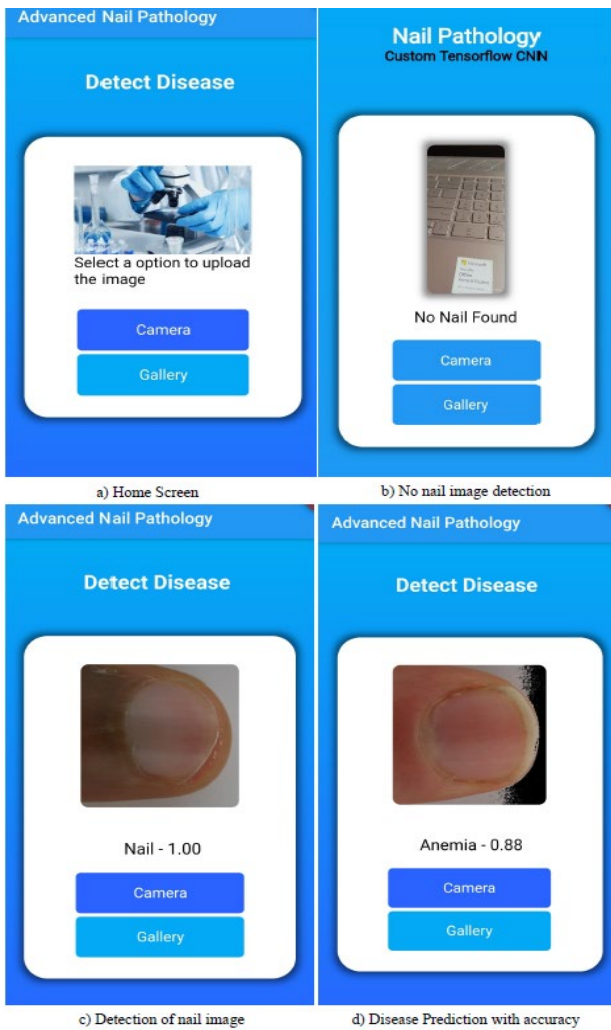


Fig. 4. Sequential Result Windows

To make this prediction again we used one more CNN model. This was done to avoid any arbitrary image as an input. Once the system identifies the input image as a nail image it proceeds to disease prediction. Our next step was to integrate the UI with the machine learning model for this the machine learning model was first converted into a TensorFlow lite (tflite) file with the help of TensorFlow lite converter. Then the tflite file was deployed in the mobile application using Flutter SDK. After that, the predicted disease is printed on the screen with the confidence level of prediction with output image as shown in Fig. 4d. According to the confidence level, users can decide whether to go with the predicted result or not.

The motive behind designing the GUI is to provide this model to each needy person. The GUI design is so simple that a layman can also use it on its own. Below is the description of the application with its real-time visuals: After the loading is done, the app's home screen appears. The home screen contains two buttons to upload the nail image as shown in Fig. 4a one for uploading the image through the gallery and another for directly capturing the image using the camera. Then the disease prediction is done by the Machine Learning model for the given image.

B. Performance Comparison of Different Classifiers

Various machine learning algorithms like RandomForest, Scalar Vector Machine (SVM), KNN [12]–[14] and CNN

were implemented using Python3(TensorFlow library) and their results were compared to get the most suitable model for prediction of the diseases. TensorFlow is an ecosystem of libraries and tools which provide more flexibility in writing code for high-level operations which are essential building blocks for an advanced neural network model. We used the clustering method in CNN for data analysis. With the help of tensor flow libraries, the process of pipelining is easily done. After comparing the results, it was observed that CNN [15] gave the best accuracy as shown in the Table. II (for training and testing same set of dataset is used for each algorithm and on the basis of the performance the best suited algorithm is used). Therefore, the CNN model was used in the application (Android). TensorFlow playground has a better visualization which helps in tracking the working of machine learning models. The ecosystem provides a Tflite library which we used for integrating the model with the user interface in Flutter mobile application as it supports Tflite models for Machine learning implementation. Fig. 5a and Fig. 5b shown defines the accuracy of each prediction made for disease, classes of disease named as healthy, skin disease, anemia, and lung disease.

TABLE II. PERFORMANCE COMPARISON OF DIFFERENT CLASSIFIER

Machine Learning Models	Training Accuracy (%)	Testing Accuracy (%)
CNN	98	75
KNN	98	63.7
RANDOM FOREST	97	54.5
SVM	95	27.3

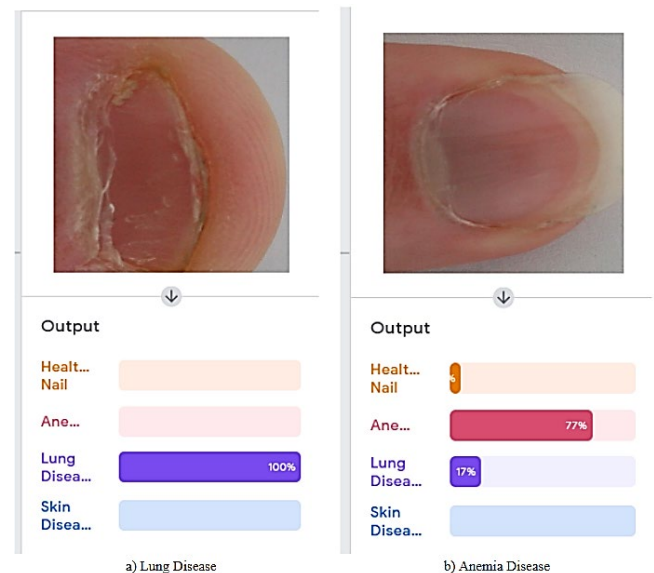


Fig. 5. Output of CNN Model for Test Images

The correlation between data is shown with the help of a confusion matrix and the loss per epoch is also mentioned. How the model classified the diseases to the true class with confidence interval is shown in Fig. 6. Confusion matrix: The Y-axis represents the true class output that was provided to

the machine and the X-axis represents the predicted class output. The Mapping is done according to the output which we get and which was decided. There are two possible outcomes of prediction yes and no. The image shows the classifier accuracy considering two random samples.

Example: If two images were tested both having ideal output healthy nail and the prediction comes out to be anemia and healthy nail, the mapping will be done as follows :

{healthy nail, healthy nail} = {2}

and

{anemia, healthy nail} = {1}

Here {2} indicates 100% probability of prediction and {1} indicates 50% probability of prediction. Accuracy and loss per epoch are shown in Fig. 7 and Fig. 8, the accuracy of the model at a chosen moment of the model is training accuracy. Increasing epoch helps in increasing the accuracy of the neural network model. The model has a testing accuracy of 75% at the 46 epoch of model training.

Confusion Matrix				
Class	Healthy Nail	Anemia	Lung Disease	Skin Disease
Healthy Nail	2	0	0	0
Anemia	1	1	0	0
Lung Disease	0	0	2	0
Skin Disease	0	1	0	1
Prediction				

Fig. 6. Confusion Matrix of CNN

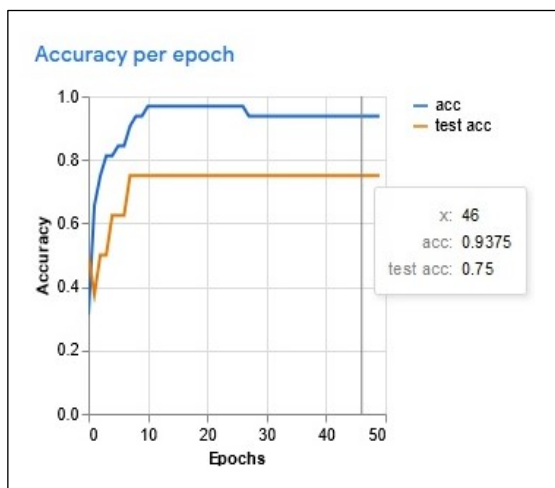


Fig. 7. Accuracy per Epoch

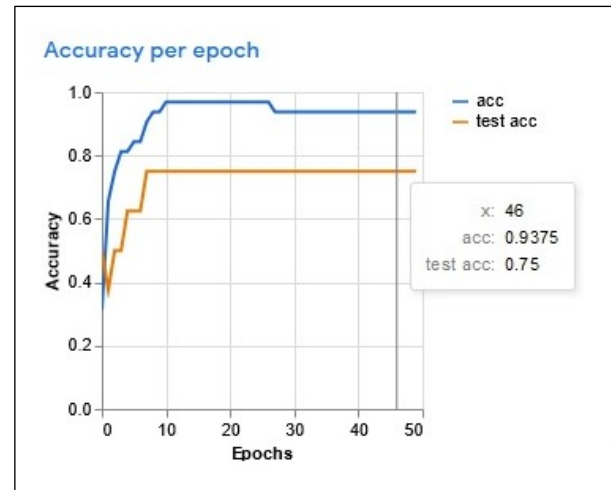


Fig. 8. Loss per Epoch

IV. CONCLUSION

The system developed can process the input image given by the user using their camera and predict the disease with the Machine Learning Model's help and show the predicted disease with its probability of occurrence. This framework requires a bit of information about the application usage and in a few seconds, the system is ready to anticipate the illness. Thus, having an idea of the occurrence of the disease before getting it worse helps the user to make a proper decision and also it helps the doctors to treat accordingly, this will save the cost and time of treatment required. The framework is made more client inviting as an application is outlined to work on smartphones, tablets, or any other smart gadgets which can provide momentary reports of disease. As the system does not store any input provided by the user so the security concerns are also very less i.e., real-time prediction is there and after that data is excised. In the future there is the scope of adding features like doctor's recommendation based on disease found and also with the collaboration of hospitals and renowned doctors we can provide direct appointments with them and the advantage of using this service here is that doctors will get a firsthand idea of a patient's situation even before starting treatment. This will save a lot of time required to do the analysis of patients and the cost of tedious tests also.

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